

LESSON 5-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Area of Parallelograms**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a parallelogram by multiplying the base by the perpendicular height.

REMEMBER

- Cut a triangle off a parallelogram and slide it over to make a rectangle: **$A = b \times h$** .
- The height h is **perpendicular** to the base. Do not use the slanted side.
- Area is in **square units** (cm², in², ft²).

WORKED EXAMPLE

Find the area: base 15 in, slanted side 8 in, height 6 in.

Step 1 Use the perpendicular height: $h = 6$ (skip the 8 in side)

Step 2 $A = b \times h = 15 \times 6$

Step 3 Multiply: $15 \times 6 = 90$

Answer **90 in²**

WORD BANK

base – the side you measure from

height – straight up from the base, at a square corner

slanted side – the tilted side; not the height

area – space inside, in square units

PRACTICE

show your work • label units

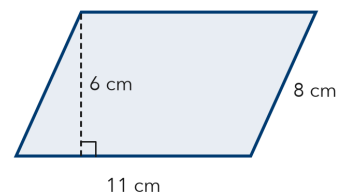
1. Find the area of the parallelogram.

Step 1 Find the base and the height (with the square corner).

Step 2 Multiply base $\times$ height. Do not use 8 cm.

Show your work.

Answer _____ cm²



2. Find the area of each parallelogram with base b and height h .

Step 1 Multiply $b \times h$ for a, b, and c.

Step 2 For d, divide the area by the height.

a. $b = 12$ ft, $h = 7$ ft _____ ft²

b. $b = 9$ m, $h = 9$ m _____ m²

c. $b = 20$ in, $h = 4$ in _____ in²

d. $A = 63$ m², $h = 7$ m; $b =$ _____ m

- 3. Parking Lot.** A parking lot has 10 spaces. Each space is a parallelogram: base 9 ft, height 18 ft. What **total** area do the 10 spaces cover?

Step 1 Find the area of one space.

Step 2 Multiply by 10.

Show your work.

Answer _____ ft²

- 4. Mural.** A mural is painted on a parallelogram-shaped wall section. The area is 108 ft² and the base is 12 ft. What is the perpendicular height?

Step 1 Area = base $\times$ height, so $108 = 12 \times$ ____.

Step 2 Divide 108 by 12.

Answer _____ ft

- 5. Find the Error.** A parallelogram has a base of 10 cm, a slanted side of 7 cm, and a height of 5 cm. Kiara says its area is 70 cm². Explain her mistake and find the correct area.

Frame Kiara used the ____ side instead of the _____. The area is ____ $\times$ ____ = ____ cm².

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Find a parallelogram shape at home or on the street (a tile, a sign, a parking space). Show someone where its base and height are.

LESSON 5-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Area of Rhombuses**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a rhombus using the base and height or the two diagonals.

REMEMBER

- A rhombus is a parallelogram, so $A = b \times h$ works.
- With the **diagonals**: $A = (d1 \times d2) \div 2$. Do not forget to divide by 2.

WORKED EXAMPLE

Find the area: diagonals 12 cm and 9 cm.

Step 1 Multiply the diagonals: $12 \times 9 = 108$

Step 2 Divide by 2: $108 \div 2 = 54$

Answer 54 cm^2

WORD BANK

rhombus – 4 sides, all the same length

diagonal – a line from corner to opposite corner

divide by 2 – take half

PRACTICE

show your work • label units

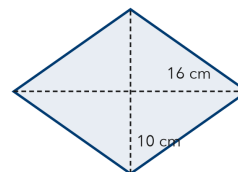
1. Find the area of the rhombus.

Step 1 Multiply the two diagonals.

Step 2 Divide by 2.

Show your work.

Answer _____ cm^2



2. Find the area of each rhombus. Choose the formula that fits.

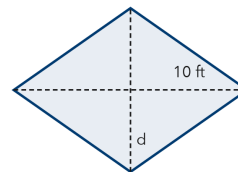
a. diagonals 14 m and 10 m _____ m^2 **b.** base 9 ft, height 7 ft _____ ft^2

3. **Window.** A rhombus window has an area of 60 ft^2 . One diagonal is 10 ft. Find the other diagonal d .

Step 1 Double the area: 2×60 .

Step 2 Divide by the diagonal you know: $\div 10$.

Answer _____ ft



4. **Stretch.** Rhombus A has diagonals of 12 ft and 8 ft. Rhombus B has a base of 8 ft and the **same area** as A. Find the height of B.

Step 1 Find the area of A: $12 \times 8 \div 2$.

Step 2 Divide that area by 8.

Answer _____ ft

- 5. Find the Error.** Ravi says a rhombus with diagonals of 18 in and 8 in has an area of 144 in². Explain his mistake and find the correct area.

Frame Ravi forgot to _____. The area is $18 \times 8 \div 2 = \underline{\hspace{1cm}}$ in².

TEST PRACTICE

choose the best answer

- 6.** A rhombus has sides of 7 m and a perpendicular height of 5 m. What is its area?

A 12 m² **B** 17.5 m² **C** 35 m² **D** 49 m²

- 7.** A rhombus has diagonals of 20 in and 9 in. What is its area?

Step 1 Multiply 20×9 .

Step 2 Divide by 2.

A 29 in² **B** 45 in² **C** 90 in² **D** 180 in²

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____